ENGINEERING

Better bug prevention — Where integration tests pay off

A look at three cycles of bug data, what they cost, and how integration tests can change the math.

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THE PROBLEM

Bugs that escape to production cost exponentially more to fix.

100×

Cost to fix a bug in production vs. at requirements stage

$2.41T

Annual cost of poor software quality in the US (CISQ, 2022)

>50%

Of bugs escape the unit-test stage and reach downstream

Sources: IBM/Capers Jones · CISQ CPSQ 2022 · NIST Planning Report 02-3

WHAT THE DATA SHOWS

Integration tests catch 27–55% of all defects before production.

Integration / Component Testing

Catches 27–55% of defects on average. The single highest-leverage test type for controller and API-level logic.

Combined layered testing

Elite teams achieve 85–95% defect removal efficiency by stacking integration, unit, and regression tests.

Static analysis adds another layer

Formal static analysis reaches 85% DRE on its own — but doesn't replace runtime integration coverage.

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Source: Capers Jones, Software Defect Removal Efficiency, 2021

OUR OWN EVIDENCE

Half our bugs over the last three cycles were IT-preventable.

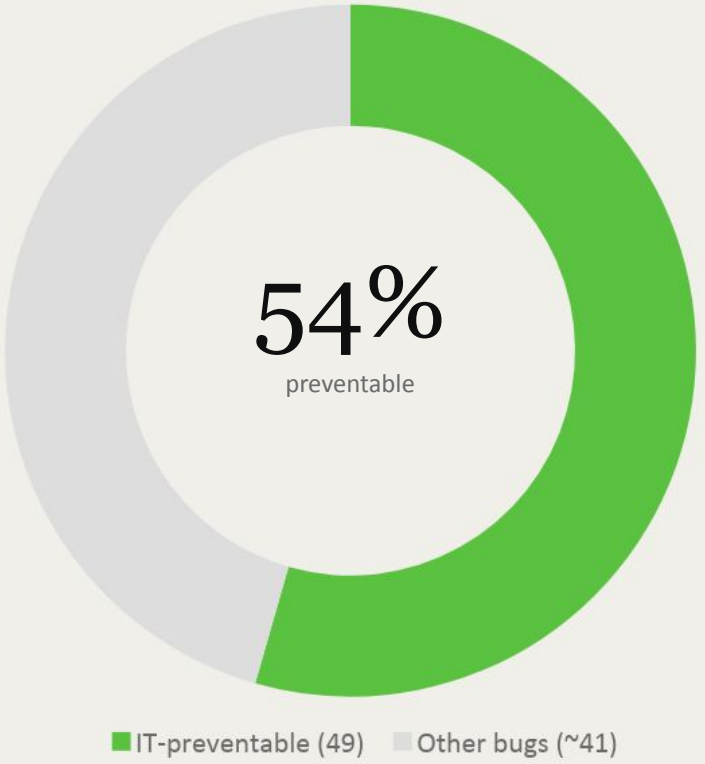
49 of ~90 bug-labeled tickets across cycles 16–18 could have been caught by an integration test using the existing harness.

5-ticket bug chain

AGI-2745 → 2799: each fix broke the next. One controller integration test would have caught the entire chain.

Shift left concept

"Shift left" is the practice of catching bugs as earlier as possible on development/deploy pipeline. As earlier they are caught, cheaper they cost



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Data: Internal Linear board analysis — Cycles 16, 17, 18 (Agentic team)

WHAT IT COST US

Those 49 bugs cost roughly $35k–$42k in dev time alone.

$28k

Dev effort to fix the 49 bugs

Standard estimate

$35–42k

All-in cost including hotfix, triage & deploy overhead

Most likely range

$5–9k

Estimated cost to write IT coverage for all 49 specs

Harness already exists

ROI on dev time alone: **3–5×** — every dollar invested in IT coverage saves three to five in bug-fixing.

Methodology: 100 story points × 6h/pt × $47/h, validated against 1,700h wall-clock at 40% active ratio.

WHAT WE BUILT

A skill that writes integration tests for us.

implement-integration-test orchestrates three specialized agents to produce coverage-complete specs against the existing IT harness.

Proven on first run

5 new tests authored, 47/47 passing on the checkout suite — including pre-existing flows.

1. assertion-checklist

Traces the flow end-to-end and enumerates every side effect — HTTP, pub/sub, DB, storage — with the fields each one touches.

2. spec-assertion-checklist

Reads the existing spec (if any) and extracts what it actually asserts, in the same schema as the source trace.

3. assertion-coverage-diff

Joins the two artifacts and classifies every row: covered, partial gap, full gap. The orchestrator only reads the diff.

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